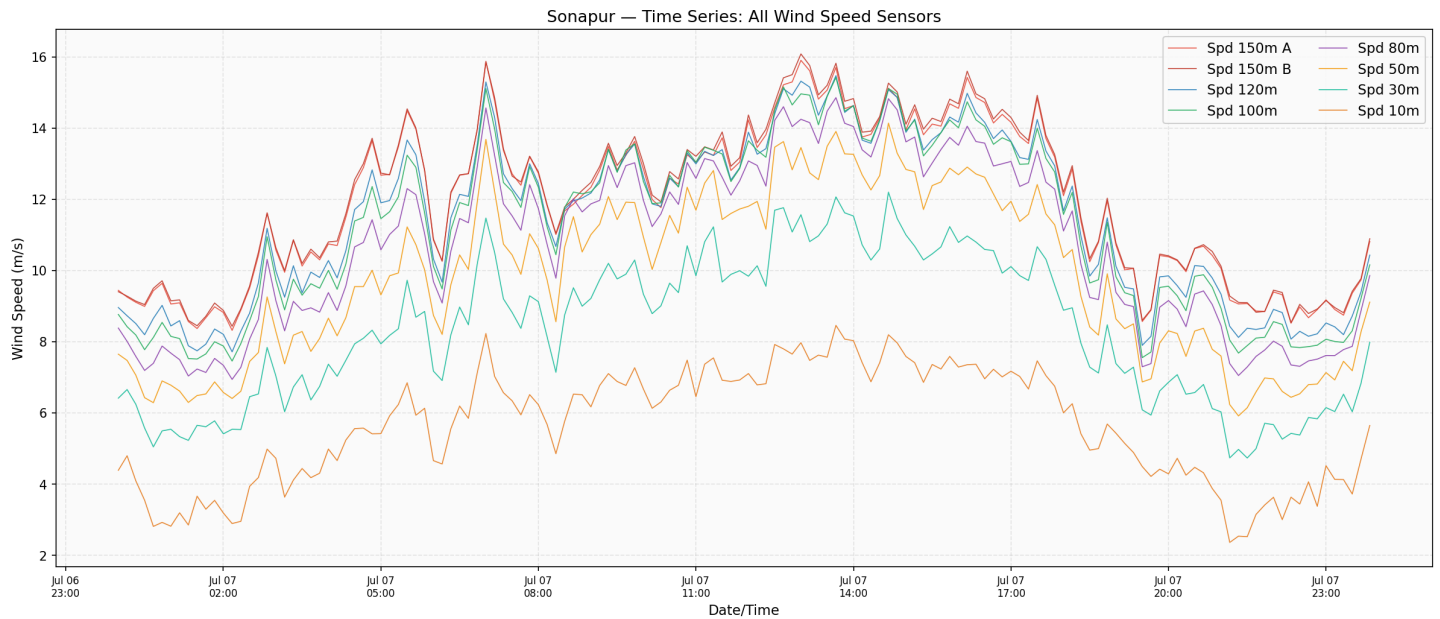


Wind Mast EDA Report

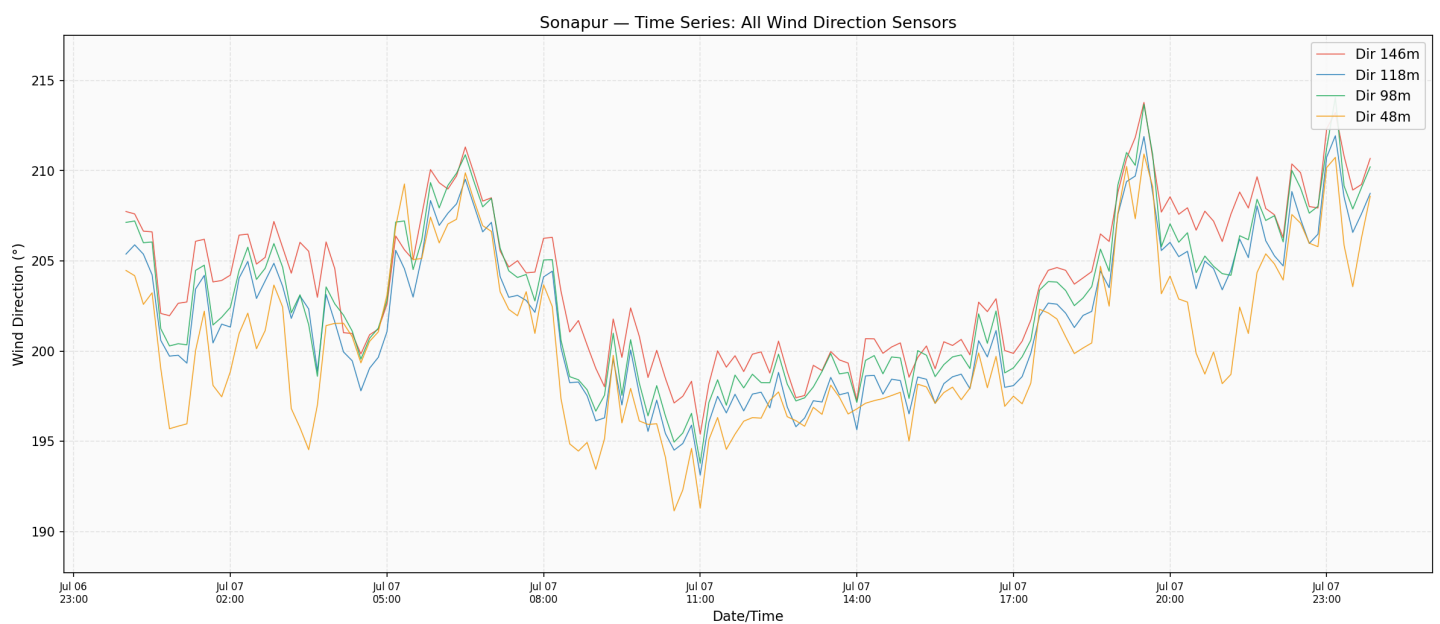
Chapter 1: Main Graphs & Summary

The charts below illustrate the time series data for wind speed, direction, temperature, humidity, pressure, and solar irradiance. The wind rose diagrams show the dominant wind directions and speed distributions for each sensor height over the analyzed period.

Source: G243136_20260707_0000.csv



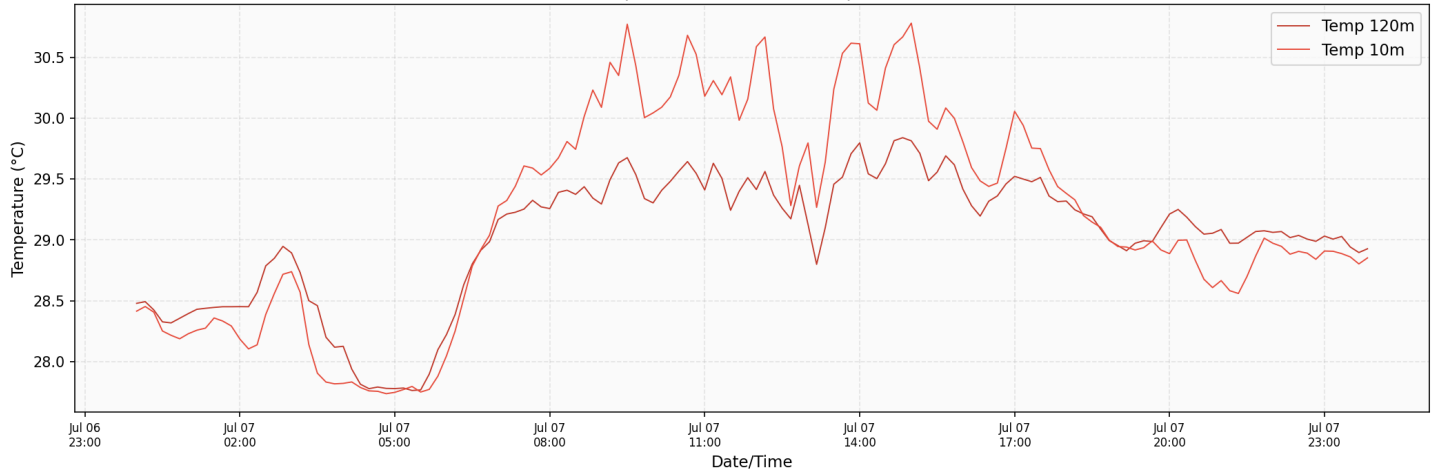
Source: G243136_20260707_0000.csv



Source: G243136_20260707_0000.csv

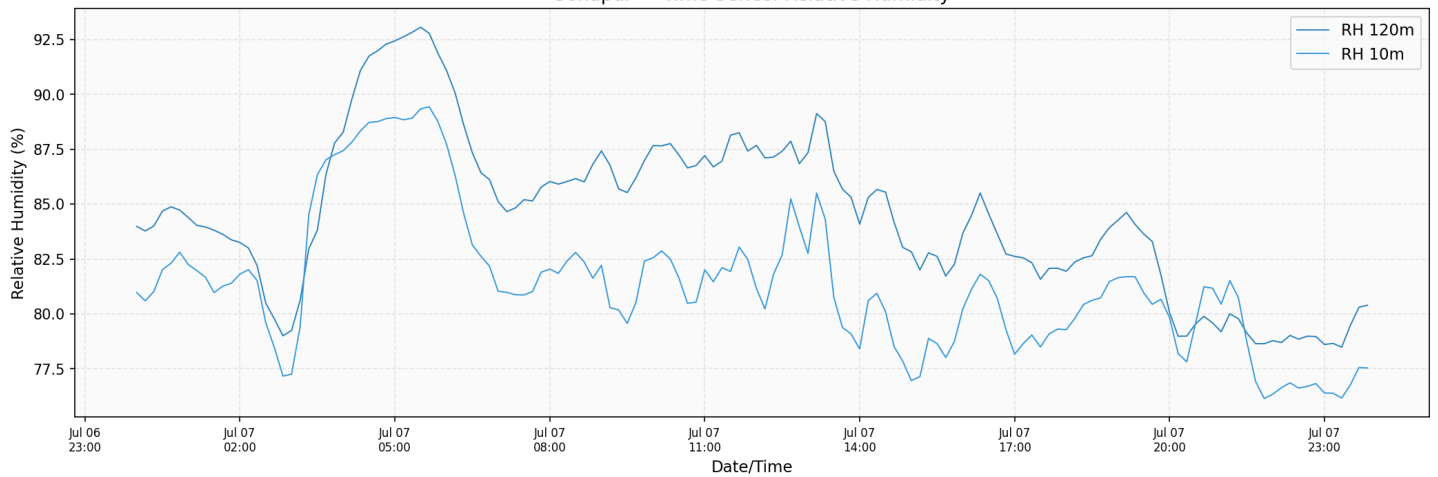
Wind Mast EDA Report

Sonapur — Time Series: Temperature



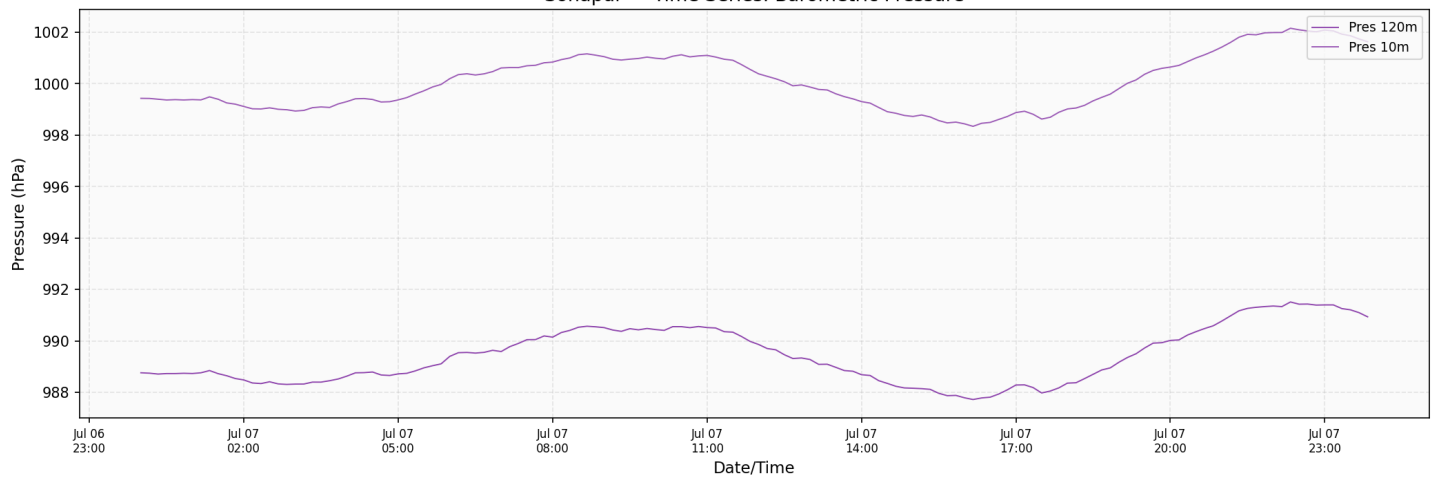
Source: G243136_20260707_0000.csv

Sonapur — Time Series: Relative Humidity



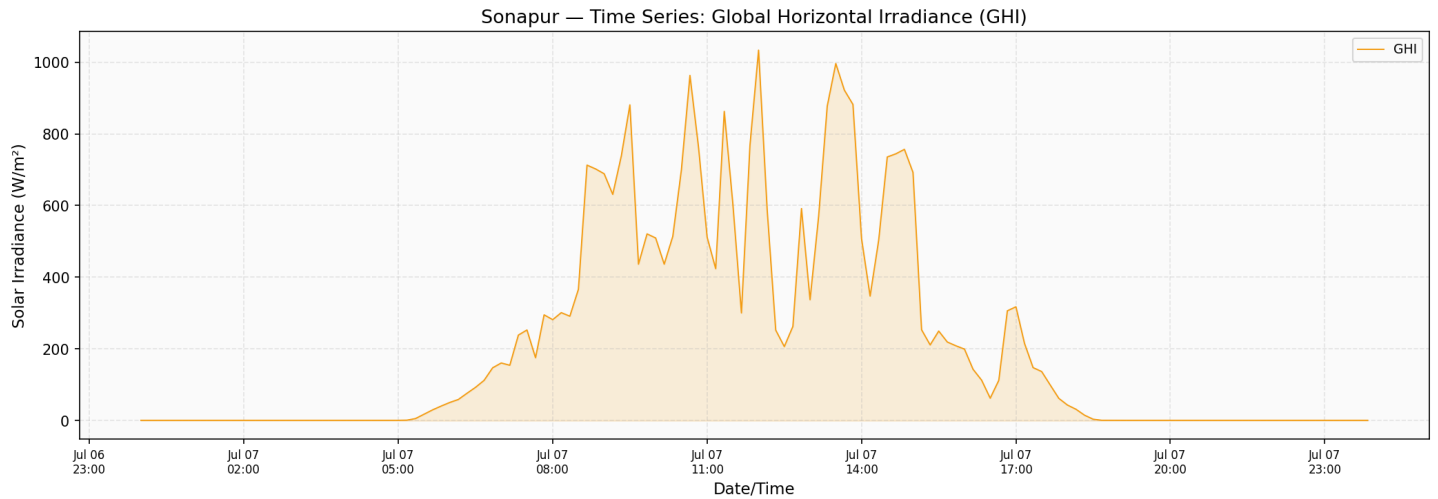
Source: G243136_20260707_0000.csv

Sonapur — Time Series: Barometric Pressure



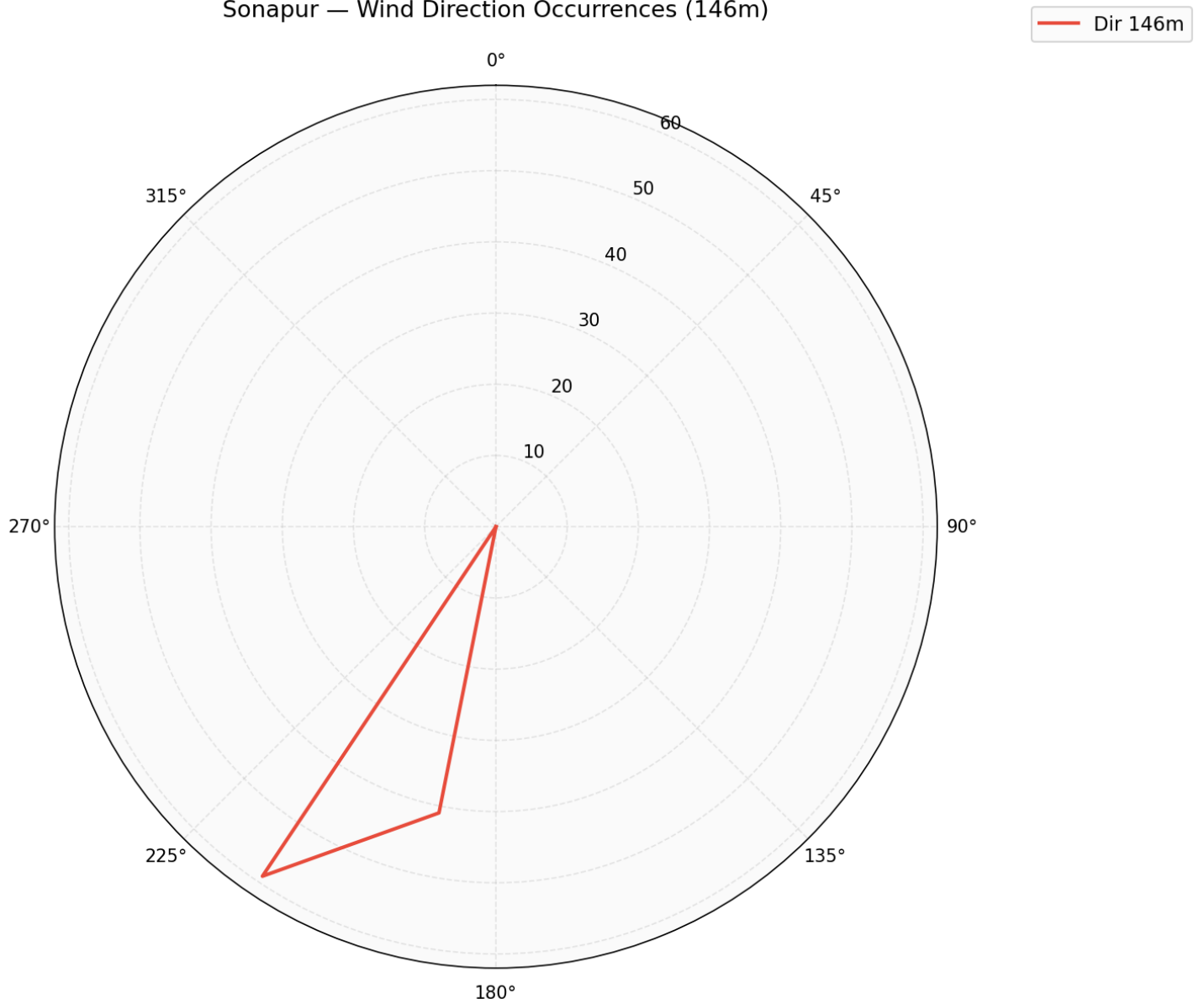
Source: G243136_20260707_0000.csv

Wind Mast EDA Report



Source: G243136_20260707_0000.csv

Sonapur — Wind Direction Occurrences (146m)

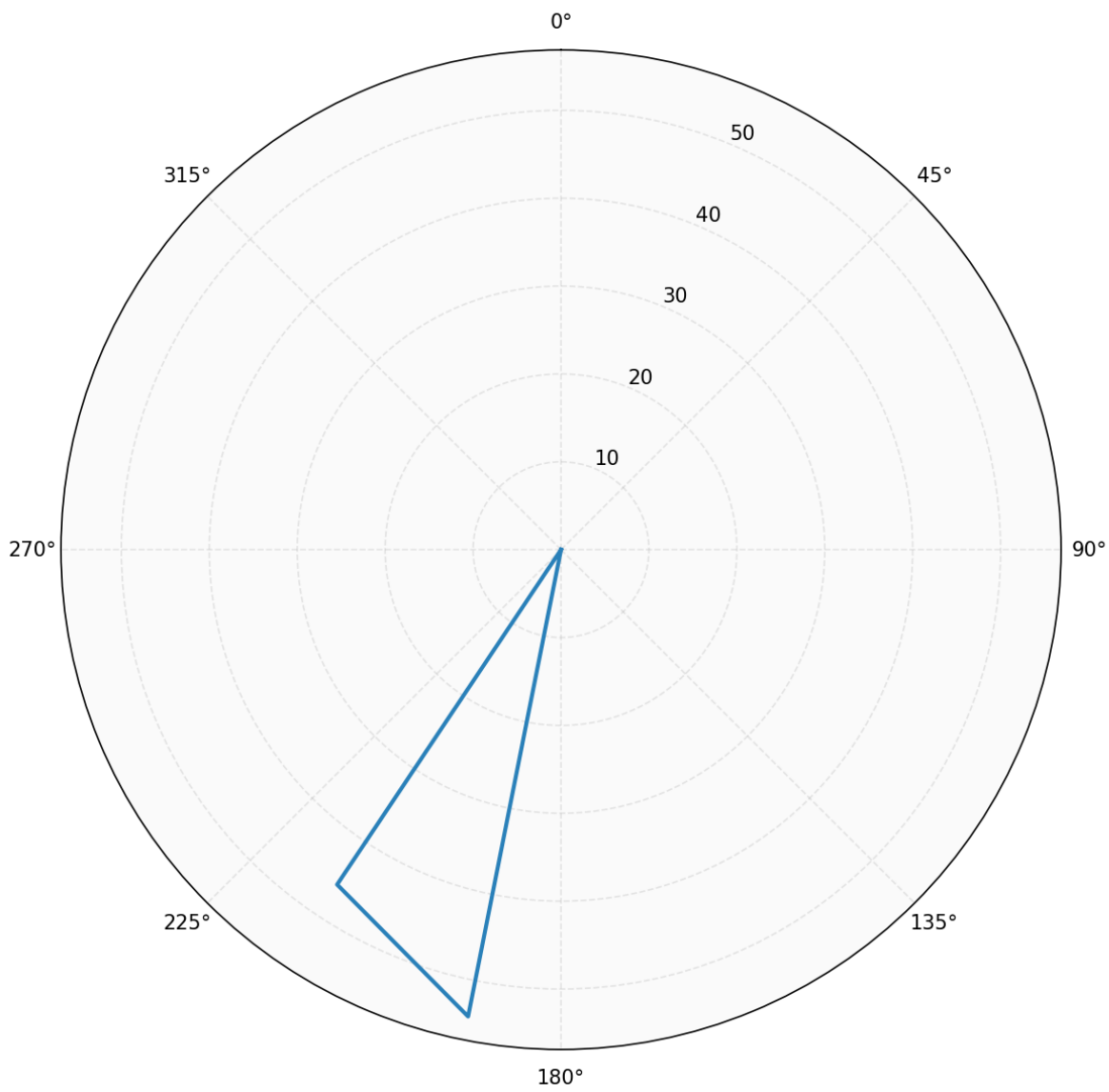


Source: G243136_20260707_0000.csv

Wind Mast EDA Report

Sonapur — Wind Direction Occurrences (118m)

Dir 118m

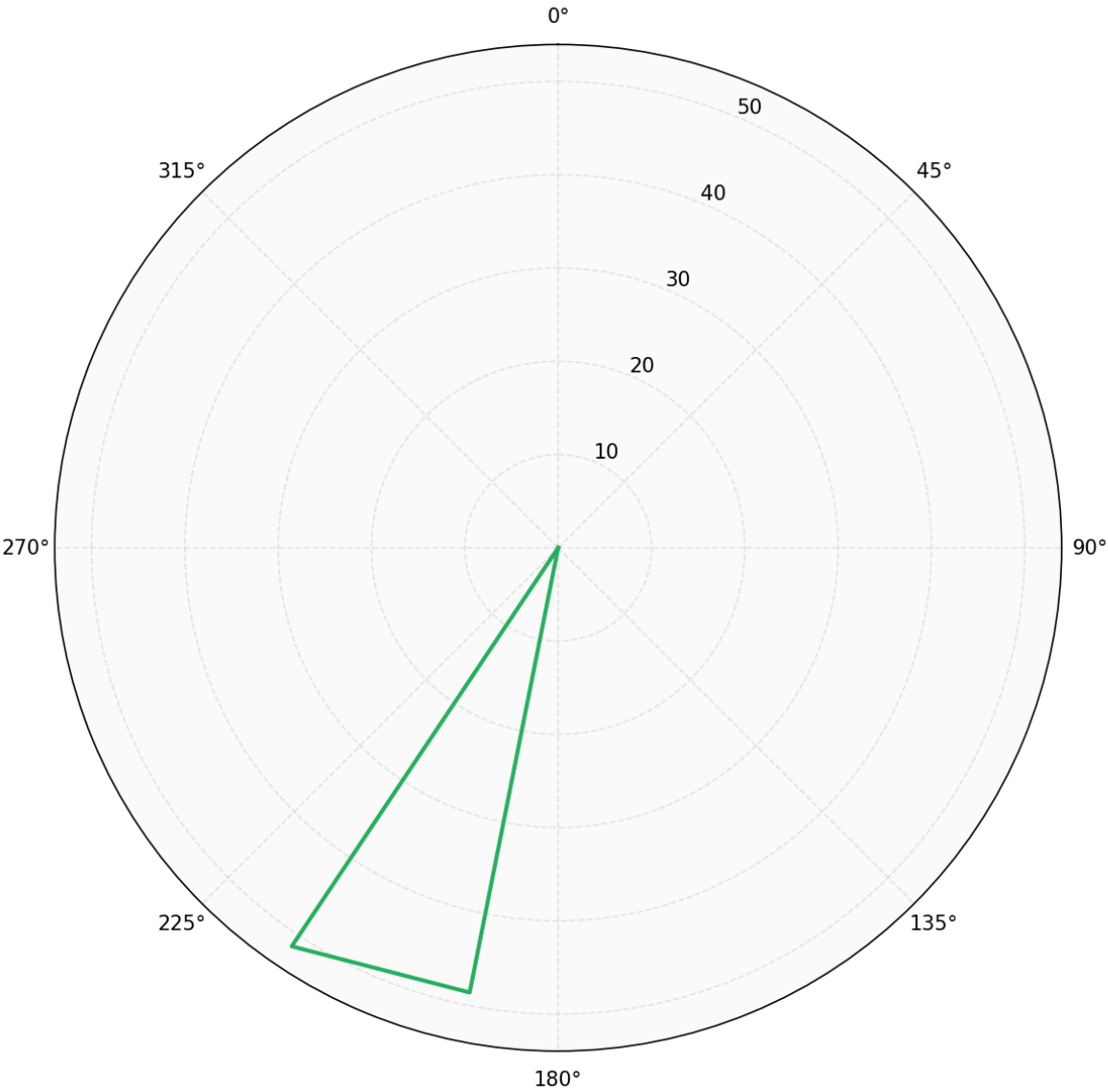


Source: G243136_20260707_0000.csv

Wind Mast EDA Report

Sonapur — Wind Direction Occurrences (98m)

Dir 98m

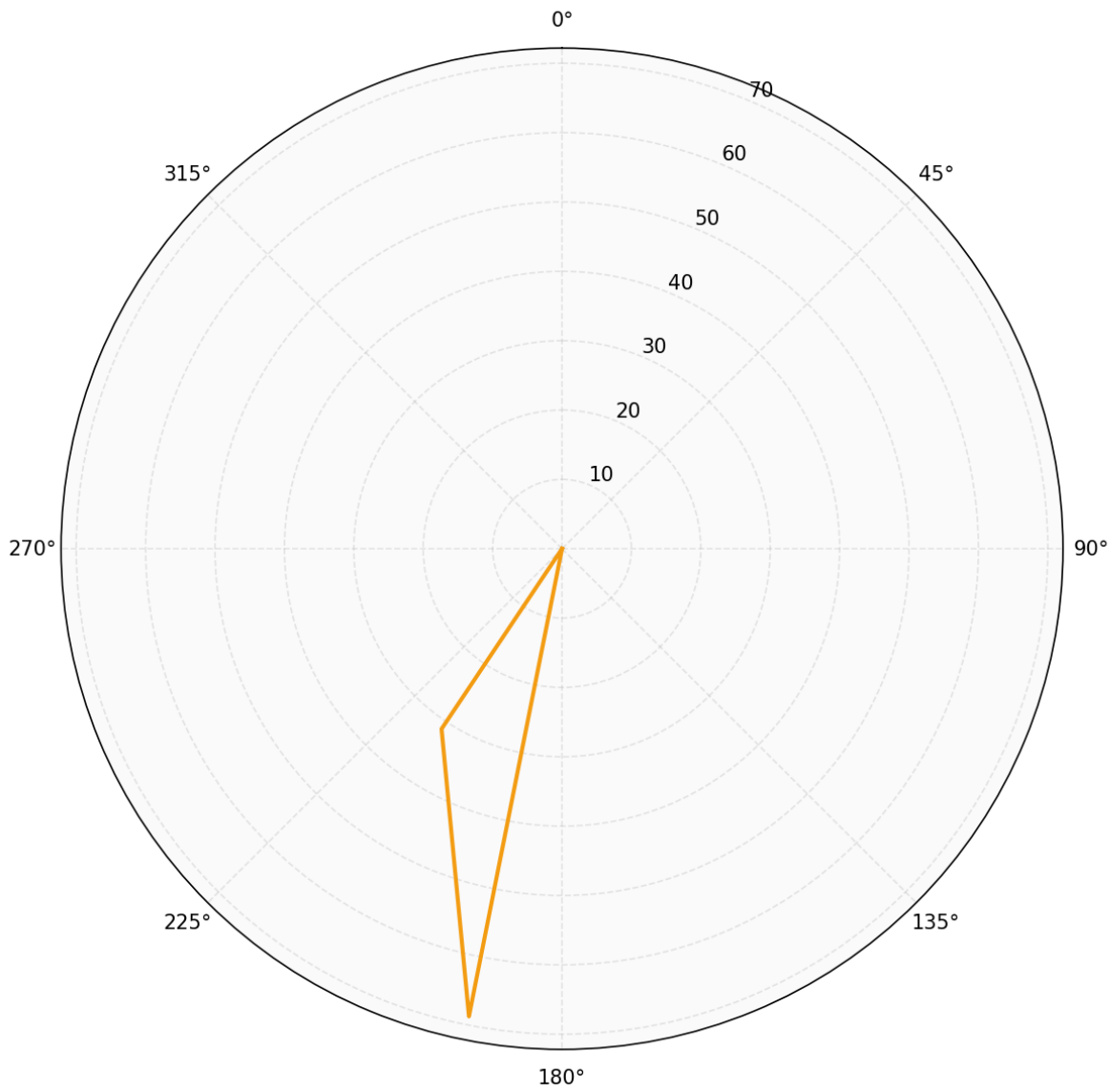


Source: G243136_20260707_0000.csv

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Sonapur — Wind Direction Occurrences (48m)

Dir 48m



Wind Mast EDA Report

Chapter 2: Correlations

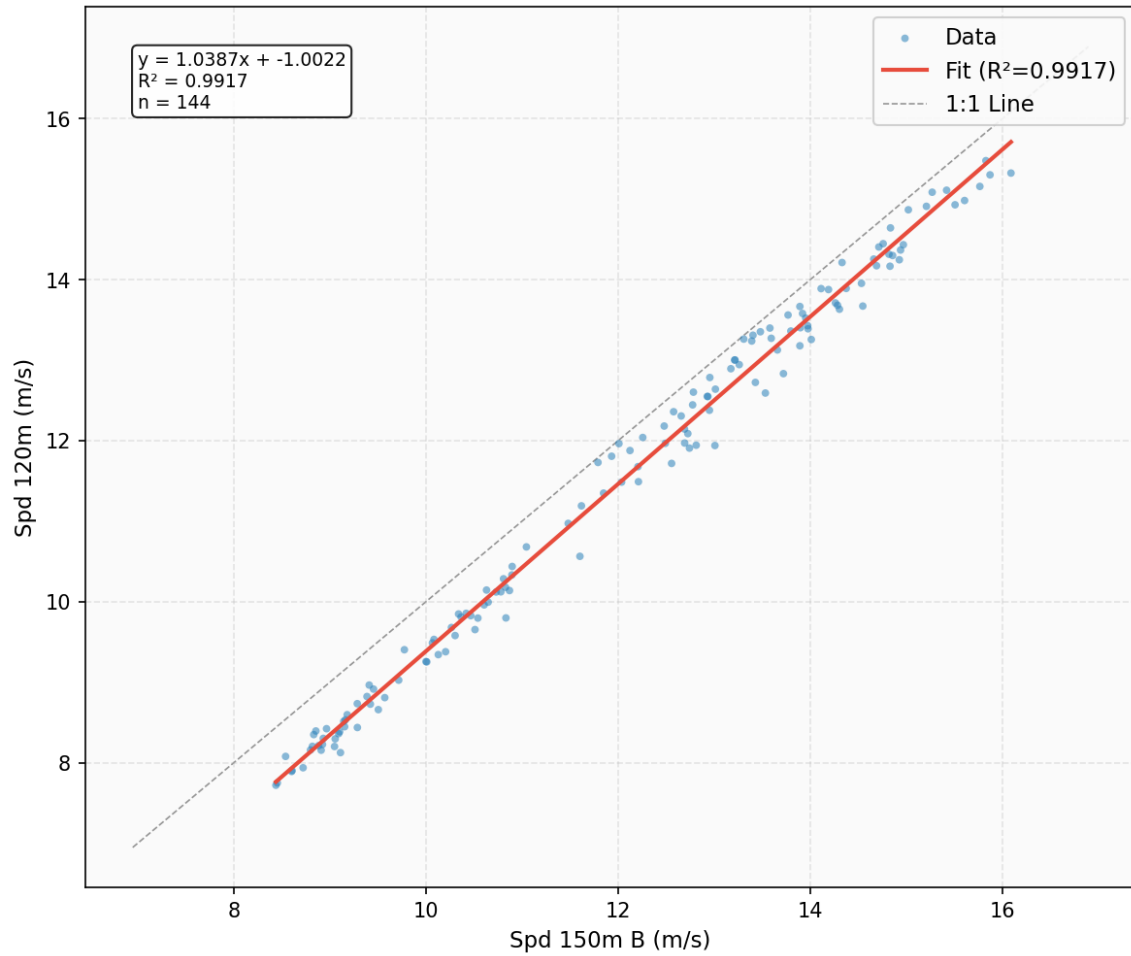
This section visualizes the coherence between different sensors at consecutive heights. High R-squared values indicate strong correlation and healthy sensor performance.

2.1 Speed Correlation

150m B vs 120m

Source: G243136_20260707_0000.csv

Sonapur — Spd 150m B vs Spd 120m

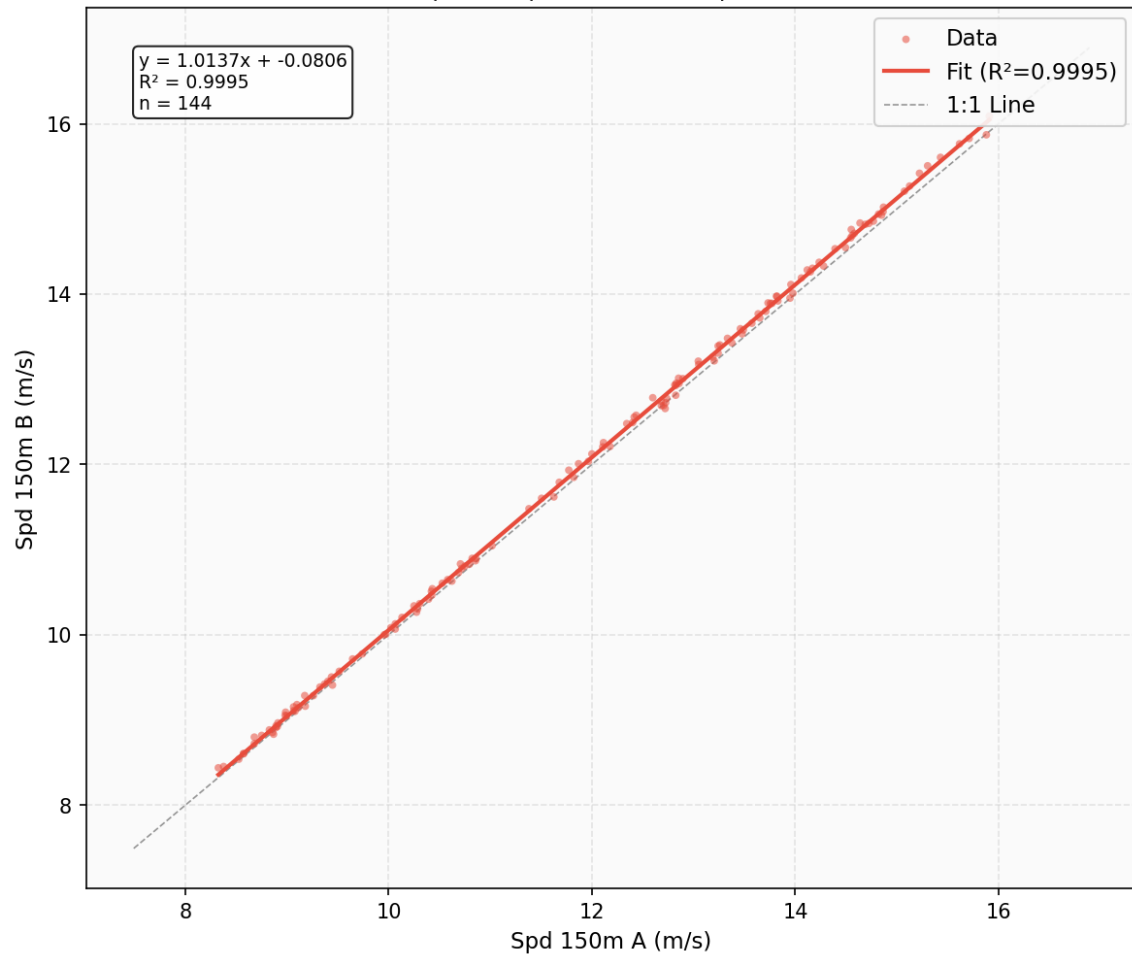


Wind Mast EDA Report

150m A vs 150m B

Source: G243136_20260707_0000.csv

Sonapur — Spd 150m A vs Spd 150m B

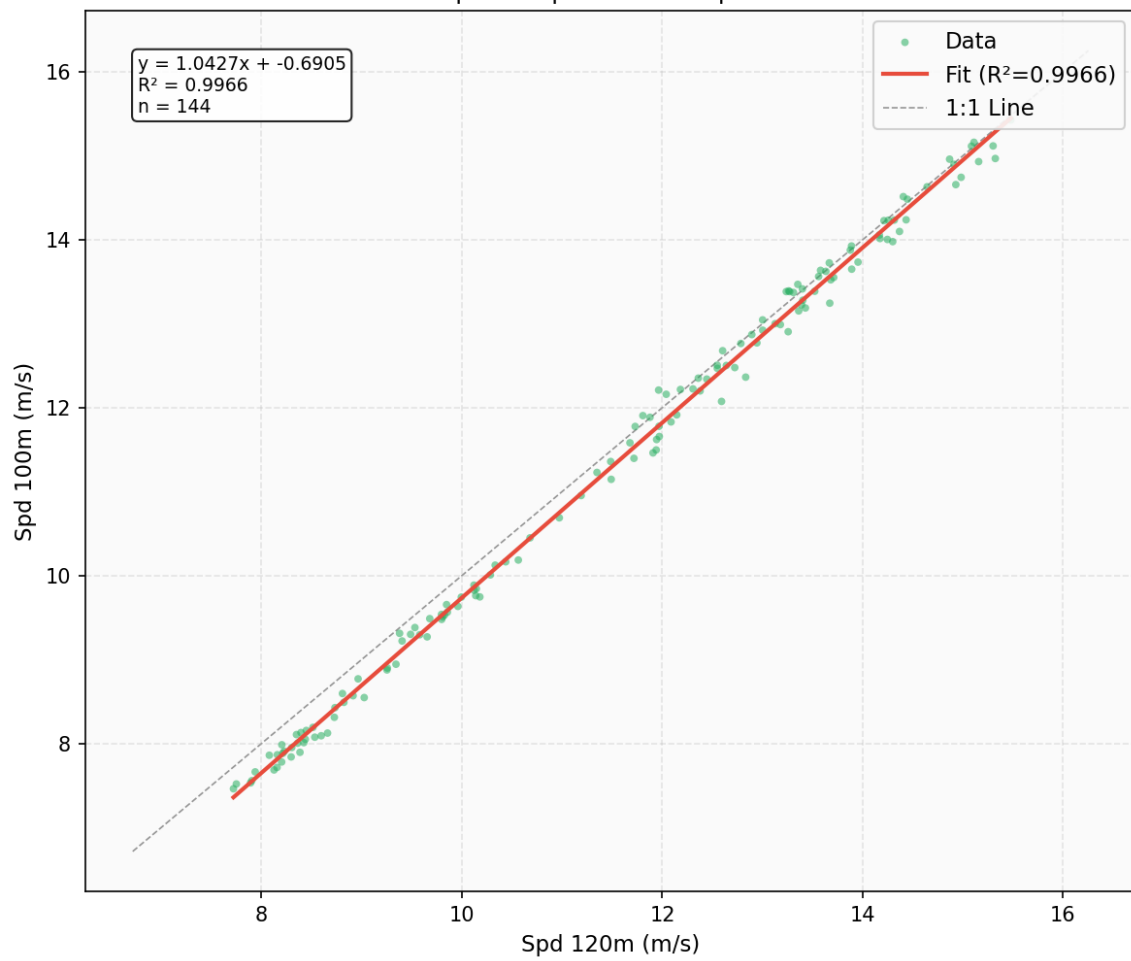


Wind Mast EDA Report

120m vs 100m

Source: G243136_20260707_0000.csv

Sonapur — Spd 120m vs Spd 100m

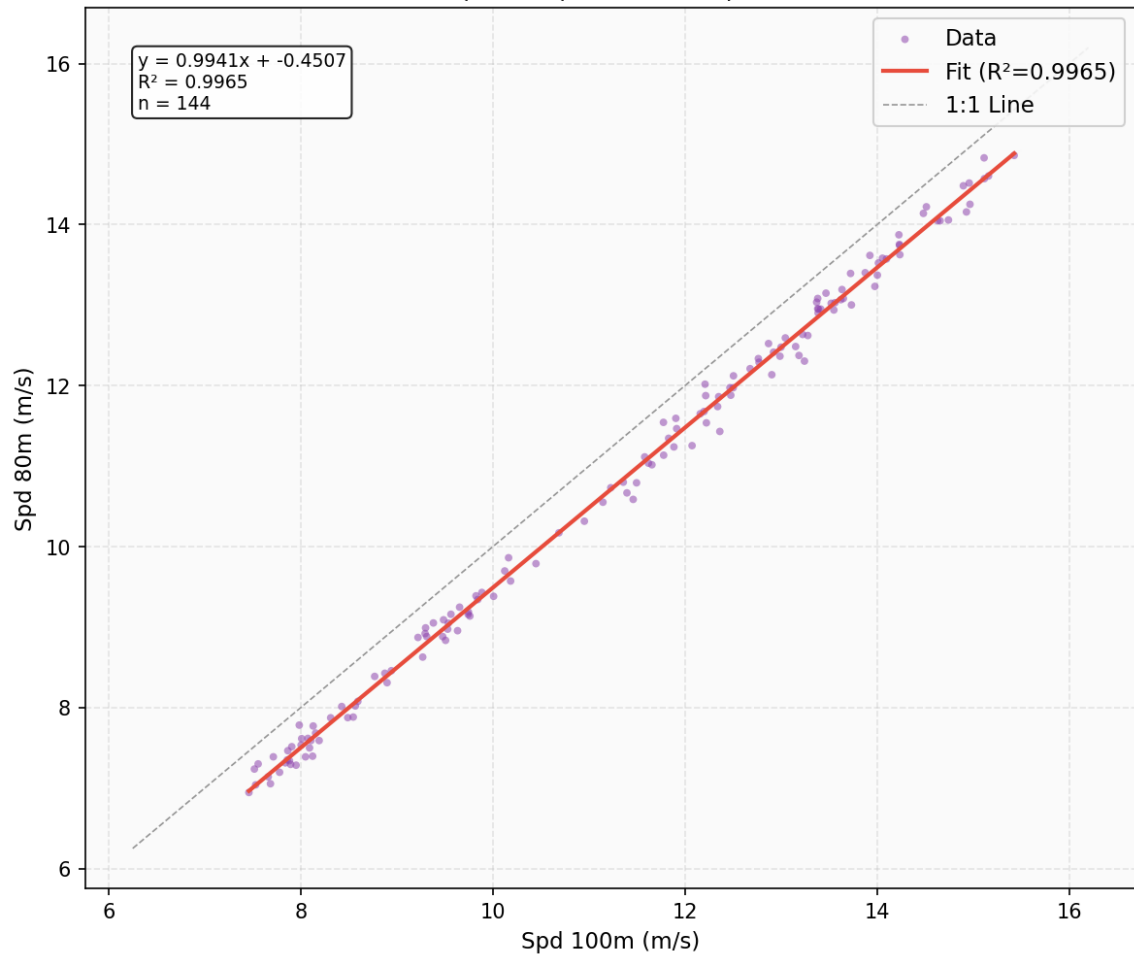


Wind Mast EDA Report

100m vs 80m

Source: G243136_20260707_0000.csv

Sonapur — Spd 100m vs Spd 80m

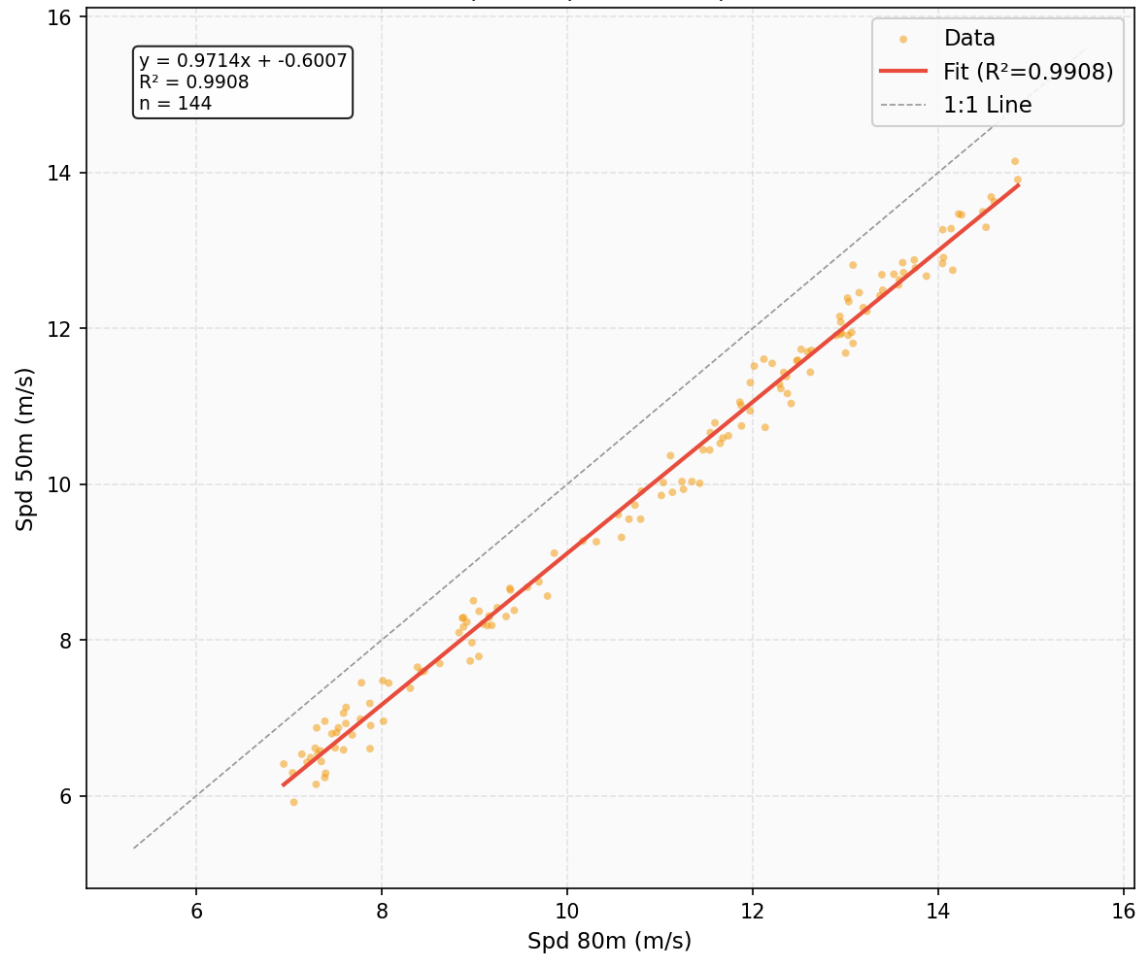


Wind Mast EDA Report

80m vs 50m

Source: G243136_20260707_0000.csv

Sonapur — Spd 80m vs Spd 50m

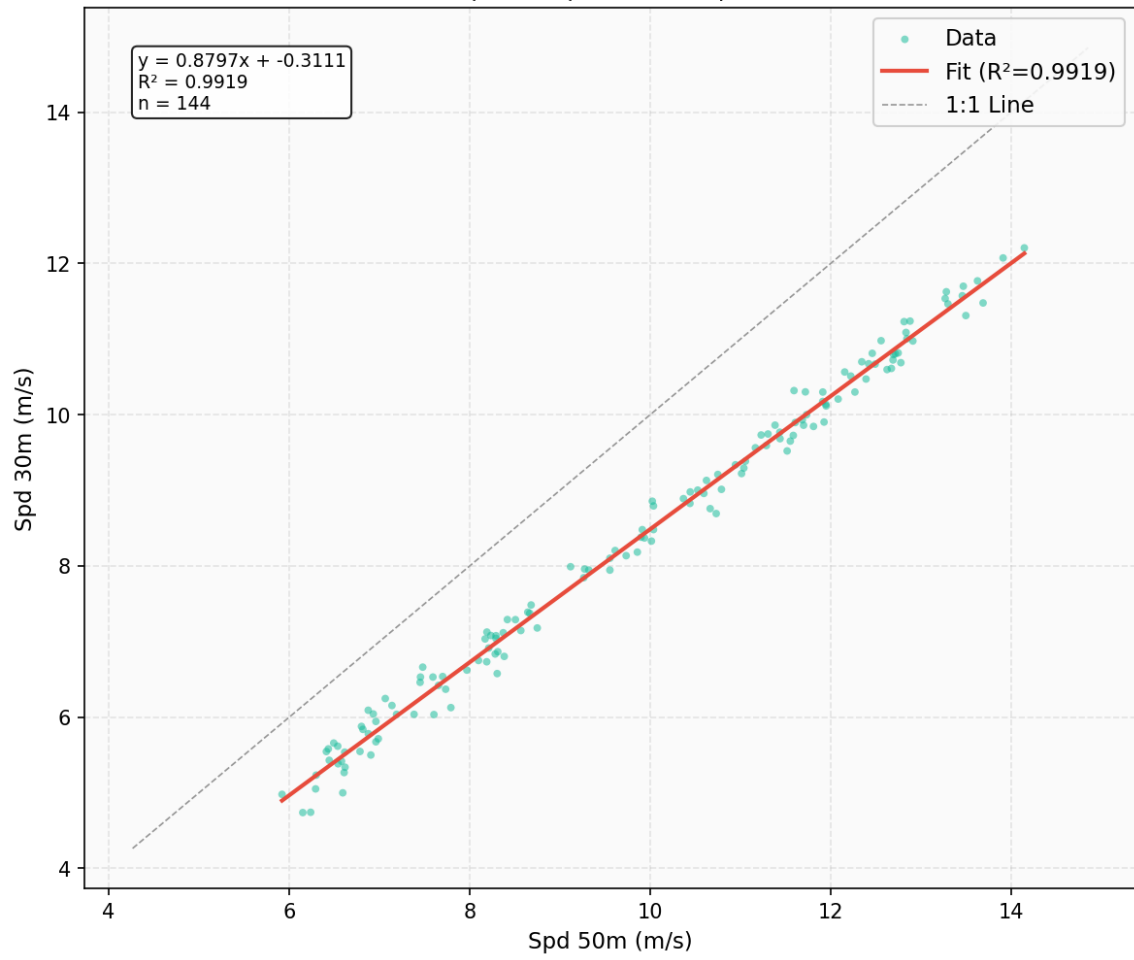


Wind Mast EDA Report

50m vs 30m

Source: G243136_20260707_0000.csv

Sonapur — Spd 50m vs Spd 30m

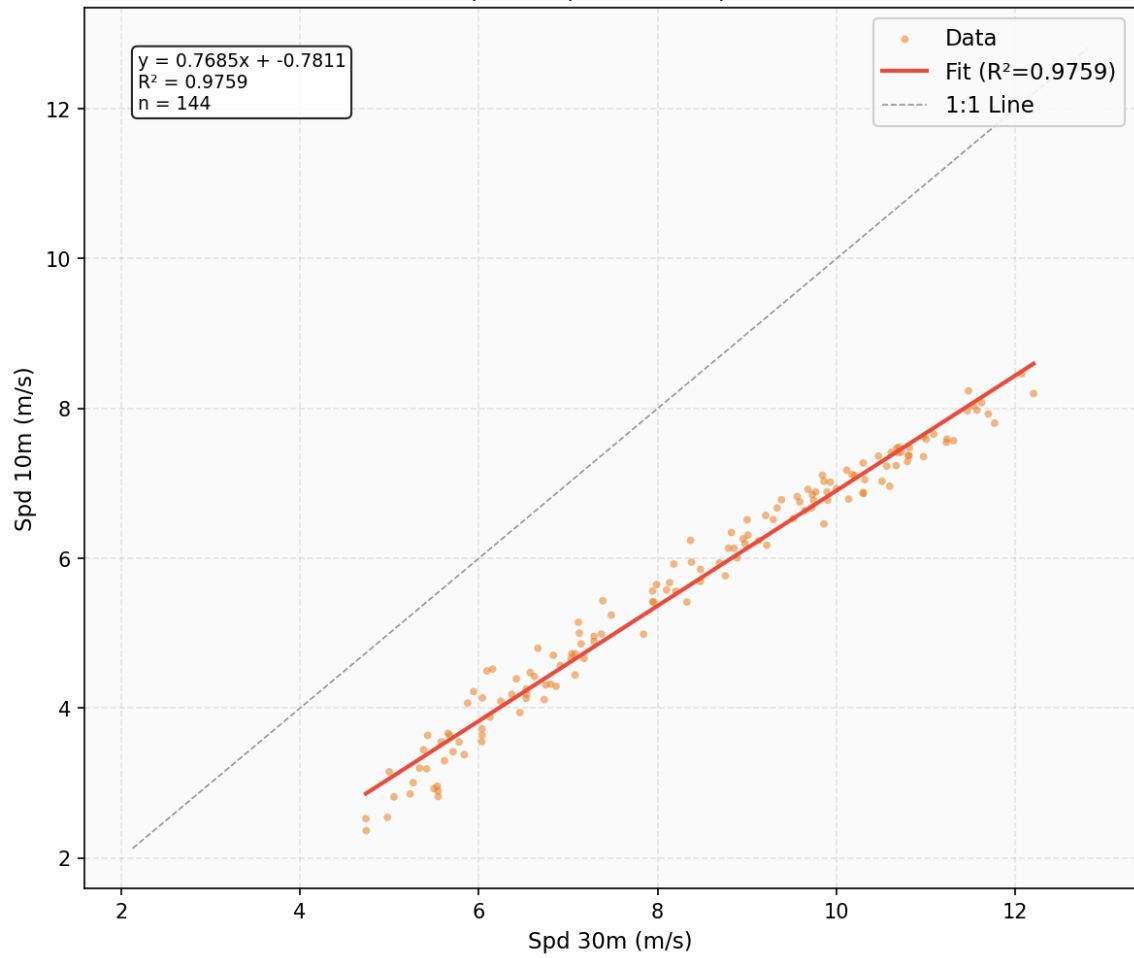


Wind Mast EDA Report

30m vs 10m

Source: G243136_20260707_0000.csv

Sonapur — Spd 30m vs Spd 10m

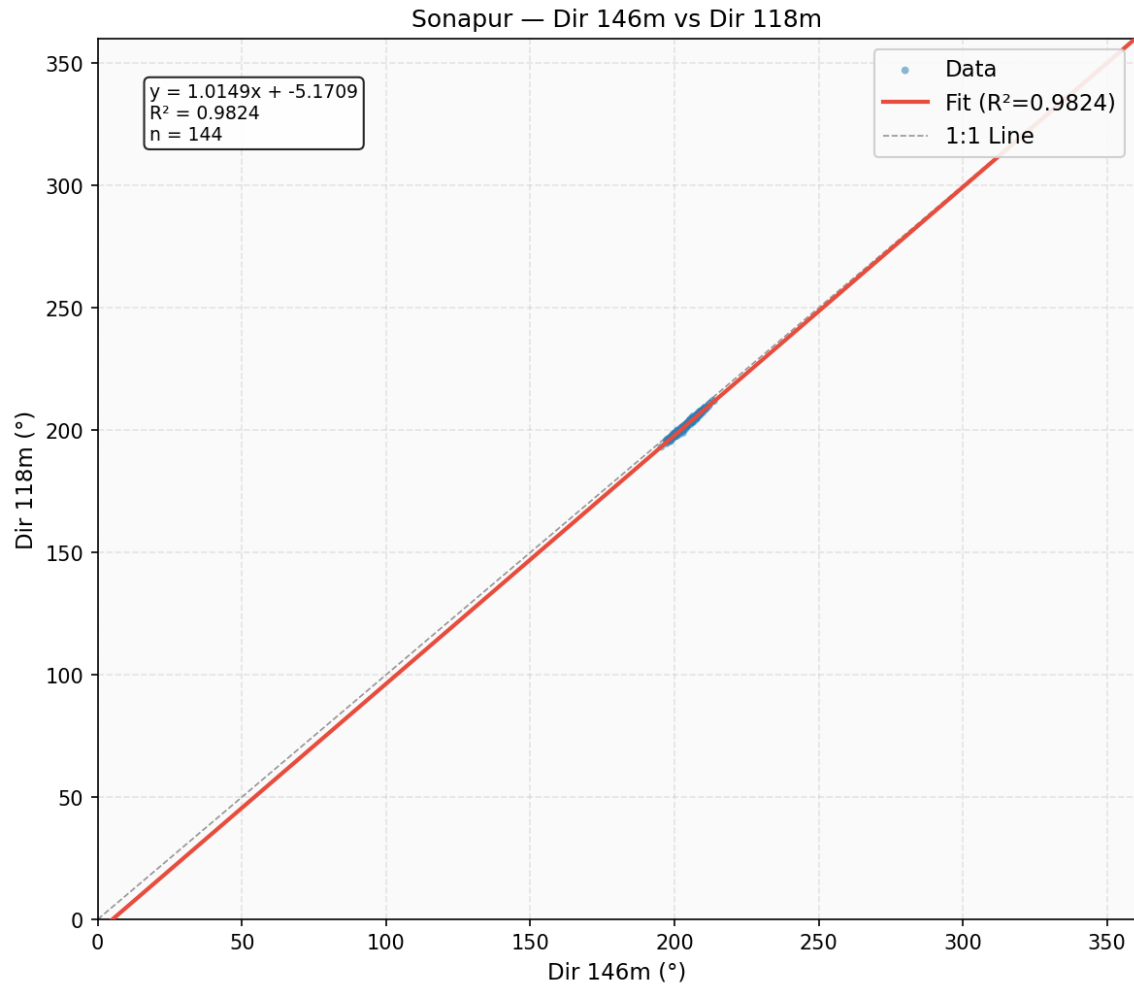


Wind Mast EDA Report

2.2 Direction Correlation

146m vs 118m

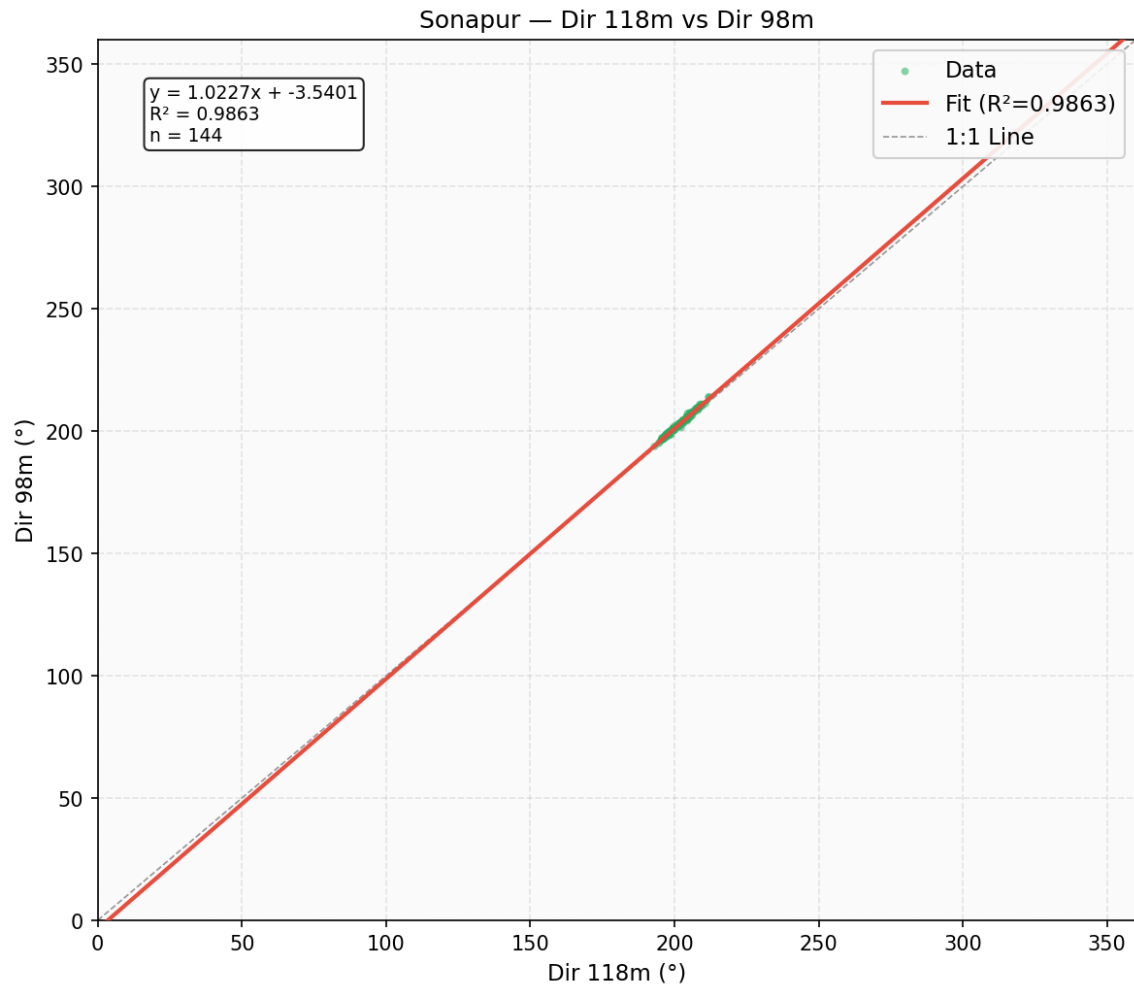
Source: G243136_20260707_0000.csv



Wind Mast EDA Report

118m vs 98m

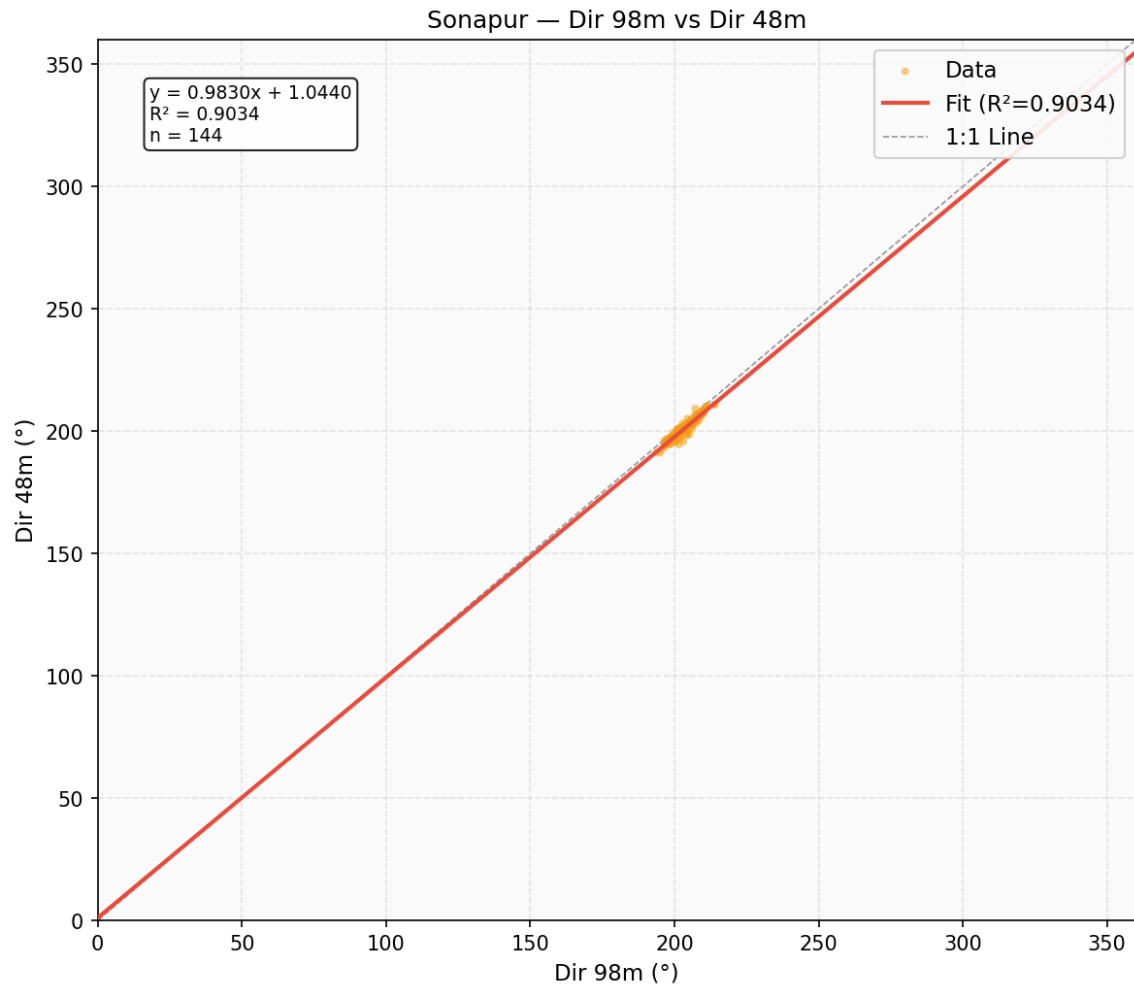
Source: G243136_20260707_0000.csv



Wind Mast EDA Report

98m vs 48m

Source: G243136_20260707_0000.csv



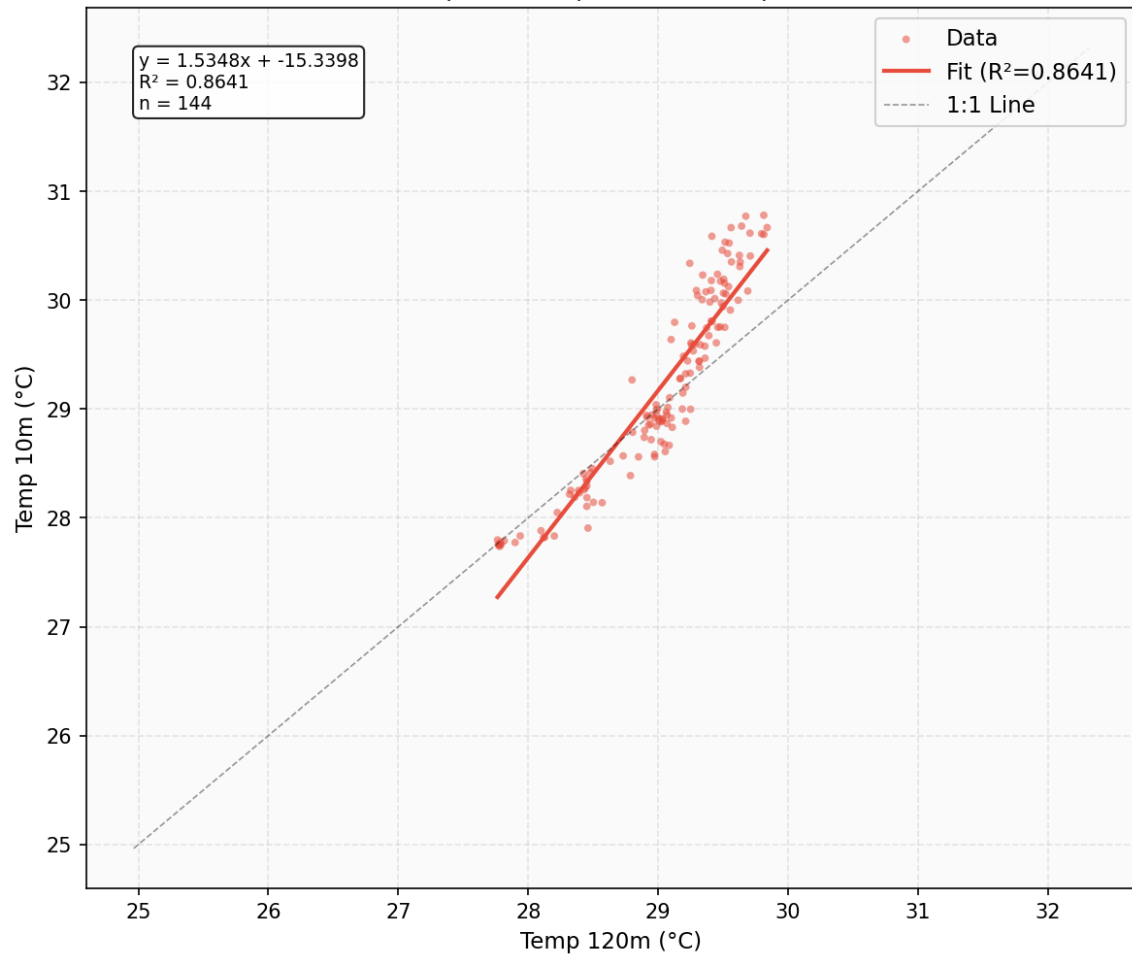
Wind Mast EDA Report

2.3 Temperature Correlation

120m vs 10m

Source: G243136_20260707_0000.csv

Sonapur — Temp 120m vs Temp 10m

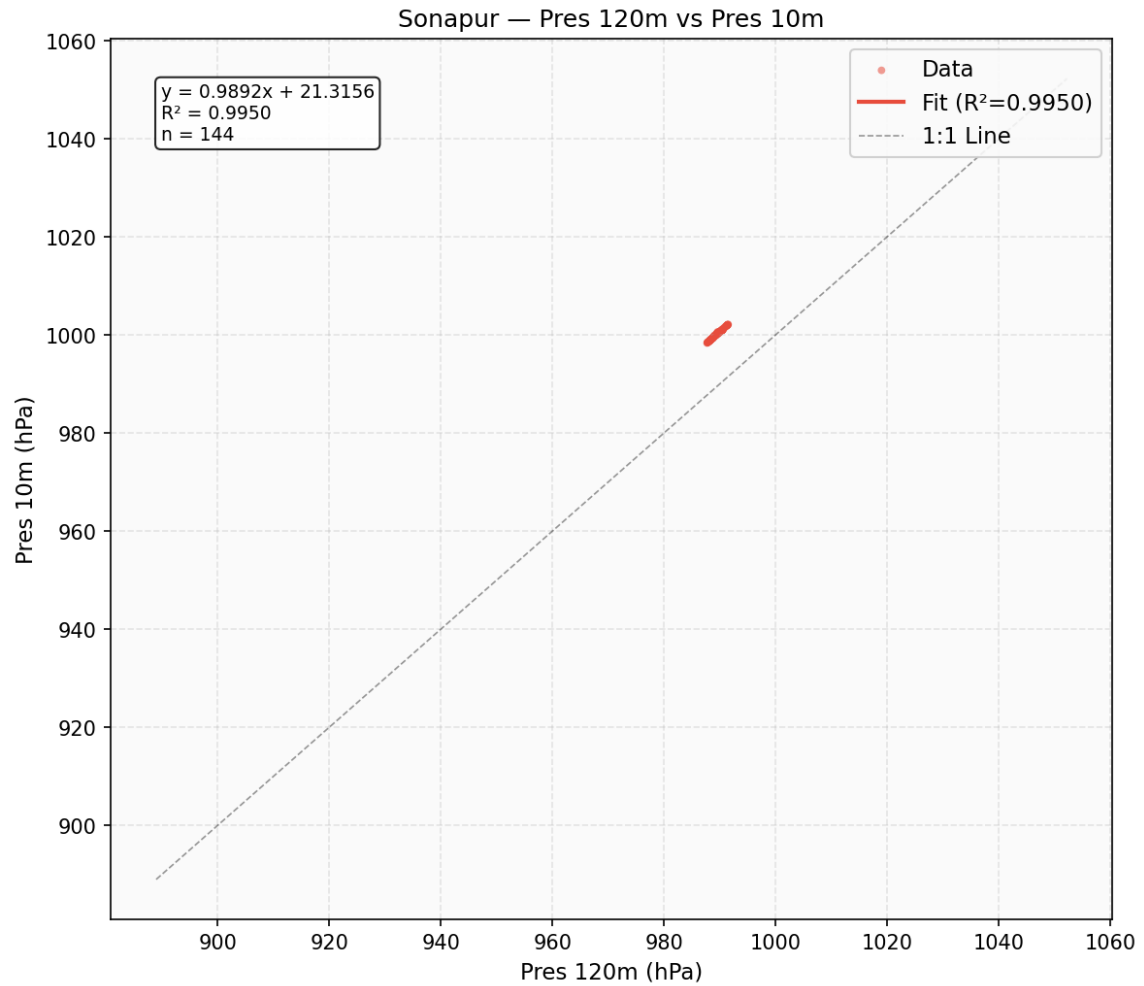


Wind Mast EDA Report

2.4 Pressure Correlation

120m vs 10m

Source: G243136_20260707_0000.csv

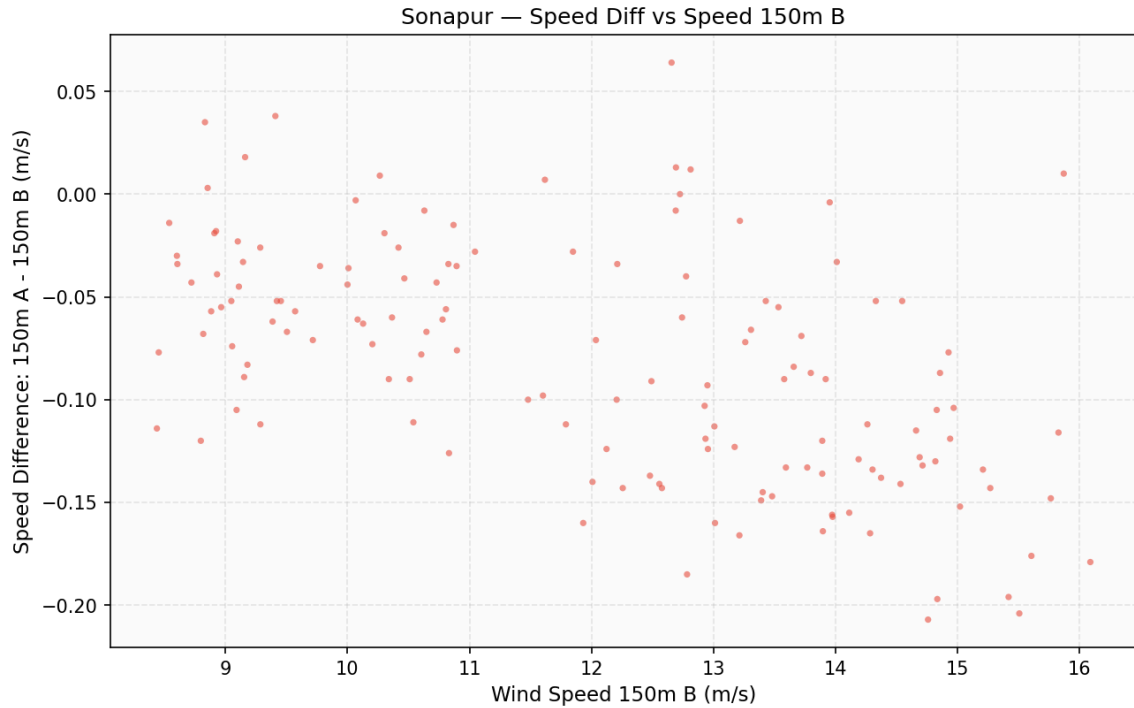


Wind Mast EDA Report

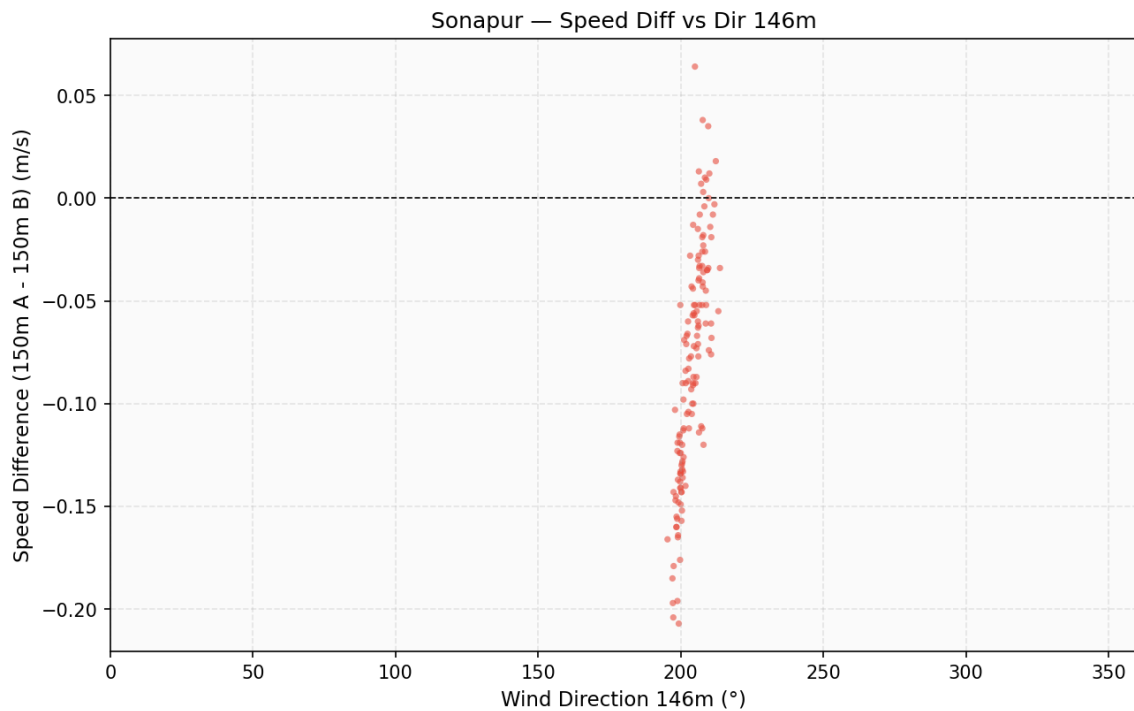
Chapter 3: Speed Difference

This section presents the speed difference between main anemometers against wind speed and wind direction.

Source: G243136_20260707_0000.csv



Source: G243136_20260707_0000.csv



Wind Mast EDA Report

Chapter 4: Data Summary

The tables below provide a statistical summary of the data, including availability (recovery rates) and average metrics.

statistical_summary.csv

S.No	Label	Units	Height	Valid Data Points	DRR (%)	Mean	Min	Max	Std. Dev.
1	Spd 150m A	m/s	150m	144	100.0	11.923	8.324	15.909	2.180
2	Spd 150m B	m/s	150m	144	100.0	12.005	8.438	16.088	2.210
3	Spd 120m	m/s	120m	144	100.0	11.468	7.720	15.477	2.305
4	Spd 100m	m/s	100m	144	100.0	11.267	7.461	15.427	2.408
5	Spd 80m	m/s	80m	144	100.0	10.750	6.946	14.860	2.398
6	Spd 50m	m/s	50m	144	100.0	9.841	5.920	14.145	2.340
7	Spd 30m	m/s	30m	144	100.0	8.346	4.738	12.205	2.067
8	Spd 10m	m/s	10m	144	100.0	5.632	2.366	8.463	1.608
9	Spd 150m A SD	m/s	150m	144	100.0	0.917	0.515	1.522	0.208
10	Spd 150m B SD	m/s	150m	144	100.0	0.924	0.531	1.527	0.208
11	Spd 120m SD	m/s	120m	144	100.0	0.918	0.581	1.543	0.191
12	Spd 100m SD	m/s	100m	144	100.0	0.949	0.612	1.564	0.196
13	Spd 80m SD	m/s	80m	144	100.0	0.964	0.605	1.644	0.199
14	Spd 50m SD	m/s	50m	144	100.0	1.146	0.584	1.835	0.277
15	Spd 30m SD	m/s	30m	144	100.0	1.385	0.772	2.251	0.357
16	Spd 10m SD	m/s	10m	144	100.0	1.554	1.076	2.201	0.245
17	Spd 150m A Gust	m/s	150m	144	100.0	14.464	11.000	19.300	2.164
18	Spd 150m B Gust	m/s	150m	144	100.0	14.542	10.200	20.100	2.227
19	Spd 120m Gust	m/s	120m	144	100.0	13.994	9.500	19.400	2.297
20	Spd 100m Gust	m/s	100m	144	100.0	13.803	9.500	18.600	2.488
21	Spd 80m Gust	m/s	80m	144	100.0	13.360	8.700	17.900	2.569
22	Spd 50m Gust	m/s	50m	144	100.0	12.758	8.000	19.300	2.770
23	Spd 30m Gust	m/s	30m	144	100.0	11.835	7.200	16.300	2.665
24	Spd 10m Gust	m/s	10m	144	100.0	9.951	5.700	14.800	2.419
25	Dir 146m	°	146m	144	100.0	204.000	195.400	213.800	4.100
26	Dir 118m	°	118m	144	100.0	201.900	193.100	211.900	4.200
27	Dir 98m	°	98m	144	100.0	202.900	193.800	214.100	4.400
28	Dir 48m	°	48m	144	100.0	200.500	191.100	210.900	4.500
29	Dir 146m SD	°	146m	144	100.0	4.000	2.200	6.800	1.100
30	Dir 118m SD	°	118m	144	100.0	4.100	2.500	7.100	1.100

... and 25 more rows.

data_availability.csv

S.No	Sensor	Height	Total Records	Valid Records	Missing Records	Data Recovery Rate (%)
1	Spd 150m A	150m	144	144	0	100.0
2	Spd 150m B	150m	144	144	0	100.0
3	Spd 120m	120m	144	144	0	100.0
4	Spd 100m	100m	144	144	0	100.0
5	Spd 80m	80m	144	144	0	100.0
6	Spd 50m	50m	144	144	0	100.0
7	Spd 30m	30m	144	144	0	100.0
8	Spd 10m	10m	144	144	0	100.0
9	Dir 146m	146m	144	144	0	100.0
10	Dir 118m	118m	144	144	0	100.0
11	Dir 98m	98m	144	144	0	100.0
12	Dir 48m	48m	144	144	0	100.0
13	Tmp 120m	120m	144	144	0	100.0
14	Tmp 10m	10m	144	144	0	100.0
15	Pres 120m	120m	144	144	0	100.0
16	Pres 10m	10m	144	144	0	100.0
17	RH 120m	120m	144	144	0	100.0
18	RH 10m	10m	144	144	0	100.0
19	GHI	NaN	144	144	0	100.0

wind_shear_summary.csv

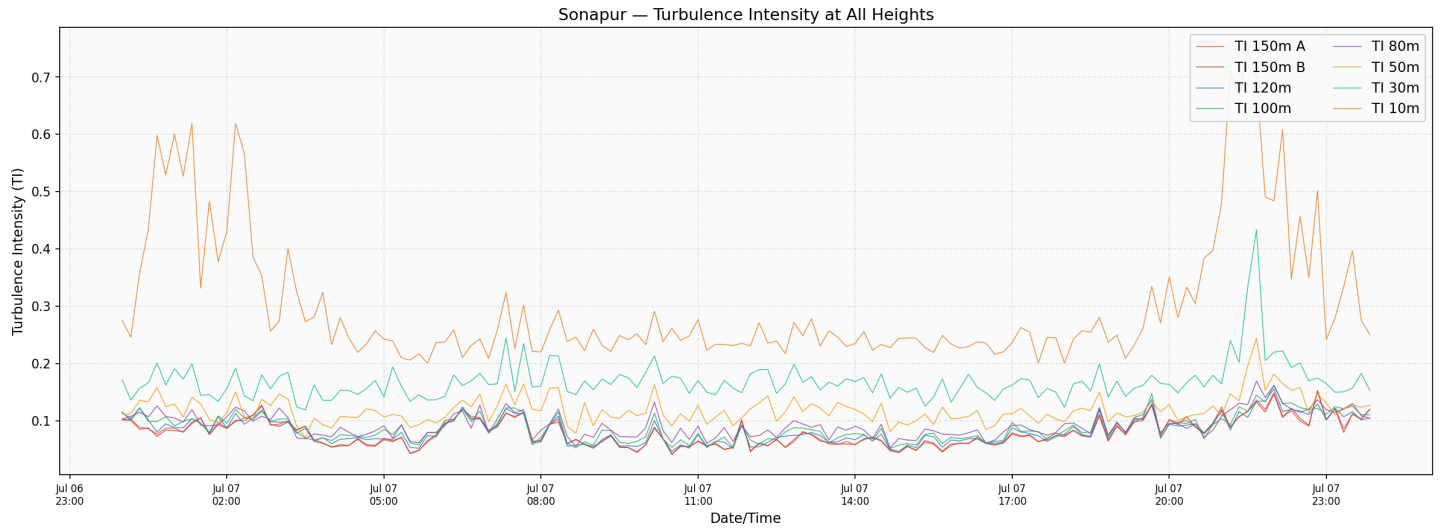
S.No	Upper Height	Lower Height	Z_upper (m)	Z_lower (m)	Alpha
1	150m	120m	150	120	0.190
2	120m	100m	120	100	0.097
3	100m	80m	100	80	0.211
4	80m	50m	80	50	0.188
5	50m	30m	50	30	0.323
6	30m	10m	30	10	0.358

Wind Mast EDA Report

Chapter 5: Other Analysis

This section contains additional analysis such as turbulence intensity distributions and the wind shear profile.

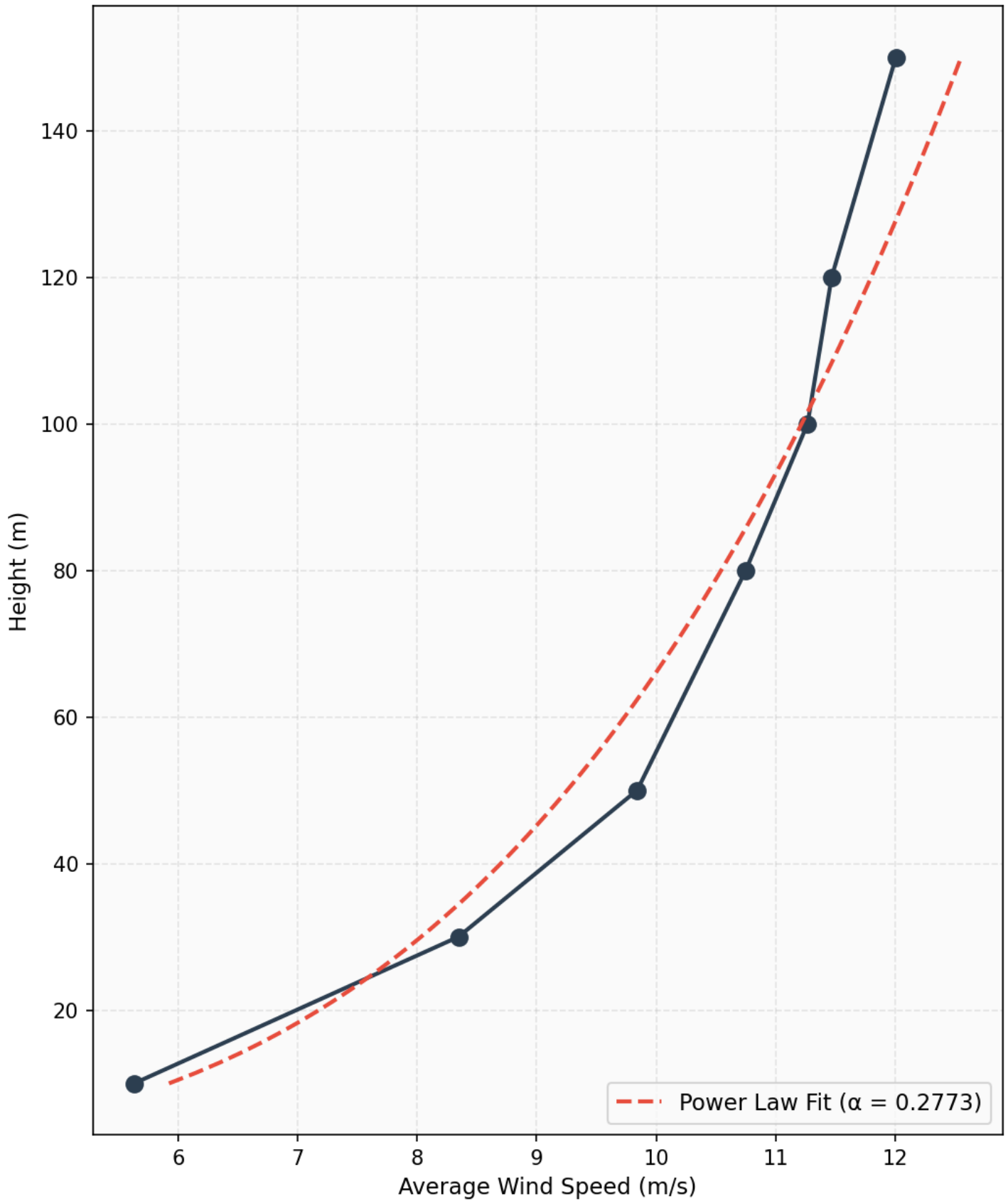
Source: G243136_20260707_0000.csv



Source: G243136_20260707_0000.csv

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Sonapur — Wind Shear Profile



Source: G243136_20260707_0000.csv

Wind Mast EDA Report

